

 UNEP	Training Session on Energy Equipment
Electrical Equipment Fans & Blowers 	Fans & Blowers Presentation from the “Energy Efficiency Guide for Industry in Asia” www.energyefficiencyasia.org   1 © UNEP 2006

TO THE TRAINER

This PowerPoint presentation can be used to train people about the basics of fans and blowers. The information on the slides is the minimum information that should be explained. The trainer notes for each slide provide more detailed information, but it is up to the trainer to decide if and how much of this information is presented also.

Additional materials that can be used for the training session are available on www.energyefficiencyasia.org under “Energy Equipment” and include:

- **Textbook chapter** on this energy equipment that forms the basis of this PowerPoint presentation but has more detailed information
- **Quiz** – ten multiple choice questions that trainees can answer after the training session
- **Workshop exercise** – a practical calculation related to this equipment
- **Option checklist** – a list of the most important options to improve energy efficiency of this equipment
- **Company case studies** – participants of past courses have given the feedback that they would like to hear about options implemented at companies for each energy equipment. More than 200 examples are available from 44 companies

in the cement, steel, chemicals, ceramics and pulp & paper sectors



Training Agenda: Fans & Blowers

Electrical Equipment
Fans & Blowers

Gerlap

Introduction

Types of fans and blowers

Assessment of fans and blowers

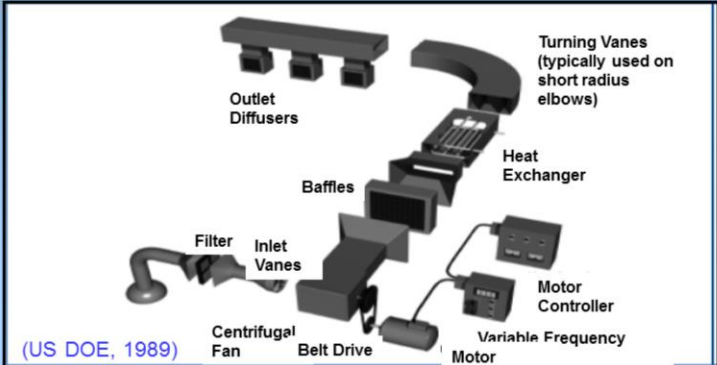
Energy efficiency opportunities

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 UNEP	Introduction
Electrical Equipment Fans & Blowers Gerlap	1. Fan components 2. System resistance 3. Fan curve 4. Operating point 5. Fan laws 3 © UNEP 2006

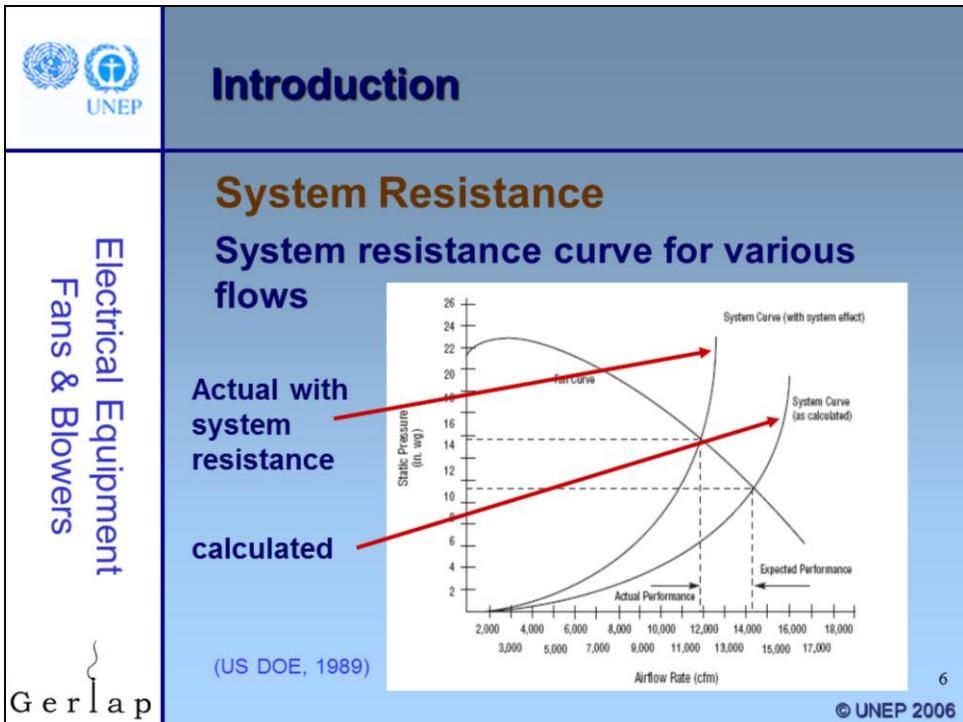
- We will first explain the fan components and go through some important theory about system and fan characteristics

 UNEP	Introduction
Electrical Equipment Fans & Blowers Gerlap	Fan Components Provide air for ventilation and industrial processes that need air flow  (US DOE, 1989) © UNEP 2006

- Most manufacturing plants use fans and blowers for ventilation and for industrial processes that need an air flow. Fan systems are essential to keep manufacturing processes working
- Fans and blowers are differentiated by the method used to move the air, and by the system pressure they must operate against.
- A typical fan system consists of a fan, an electric motor, a drive system, ducts or piping, flow control devices, and air conditioning equipment (filters, cooling coils, heat exchangers, etc.)

 UNEP	Introduction
Electrical Equipment Fans & Blowers Gerlap	System Resistance • Sum of static pressure losses in system • Configuration of ducts, pickups, elbows • Pressure drop across equipment • Increases with square of air volume • Long narrow ducts, many bends: more resistance • Large ducts, few bends: less resistance 5 © UNEP 2006

- The term “system resistance” is used when referring to the static pressure. The system resistance is the sum of static pressure losses in the system. The system resistance is a function of the configuration of ducts, pickups, elbows and the pressure drops across equipment, for example bag filter or cyclone.
- The system resistance varies with the square of the volume of air flowing through the system. For a given volume of air, the fan in a system with narrow ducts and multiple short radius elbows is going to have to work harder to overcome a greater system resistance than it would in a system with larger ducts and a minimum number of long radius turns.



- To determine what volume the fan will produce, it is therefore necessary to know the system resistance characteristics. In existing systems, the system resistance can be measured. In systems that have been designed, but not built, the system resistance must be calculated.
- Typically a system resistance curve (see Figure) is generated with for various flow rates on the x-axis and the associated resistance on the y-axis.
- The calculated or expected system curve in this figure has higher flow rates for a given flow rate than the actual system curve. This is due to the system resistance.

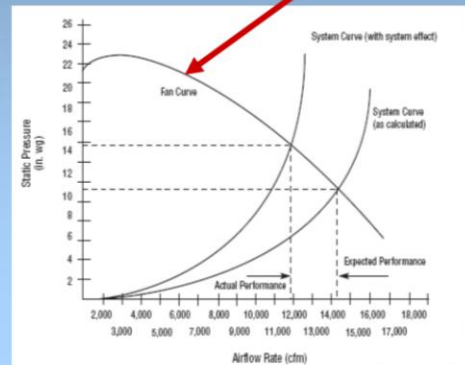
Introduction

Fan Curve

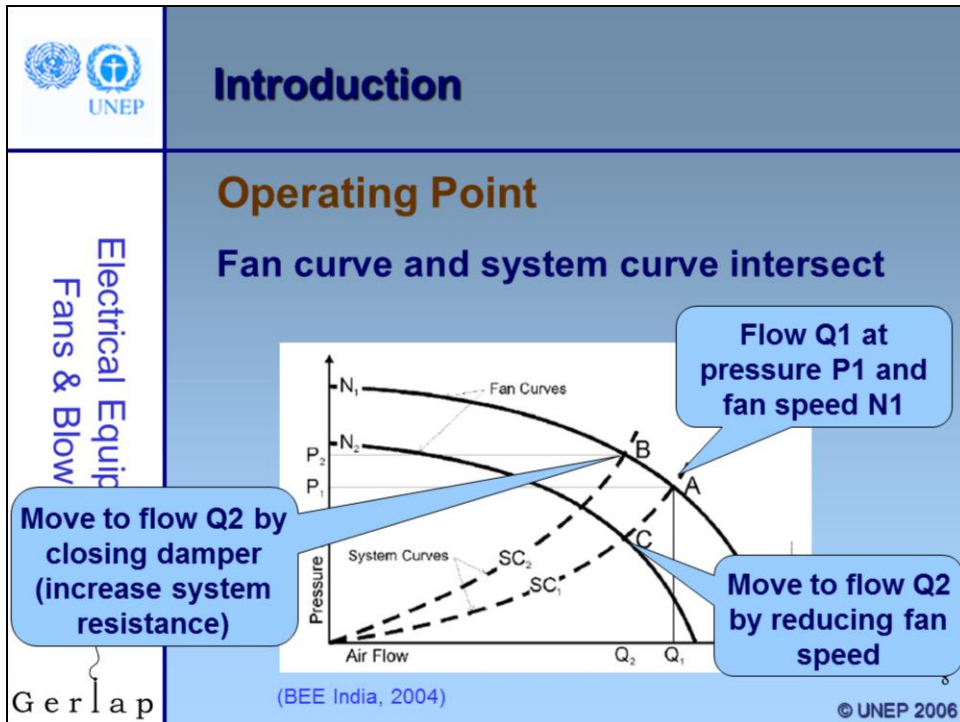
Performance curve of fan under specific conditions

- Fan volume
- System static pressure
- Fan speed
- Brake horsepower

(US DOE, 1989)

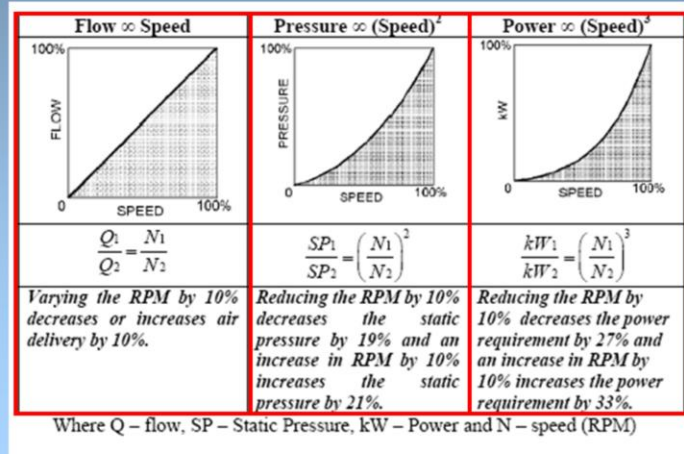


- Fan characteristics can be represented in form of fan curve(s). The fan curve is a performance curve for the particular fan under a specific set of conditions, usually including: fan volume, system static pressure, fan speed, and brake horsepower required to drive the fan under the stated conditions.



- This system curve can then be plotted on the fan curve to show the fan's actual **operating point** at "A" where the two curves (N1 and SC1) intersect. The fan's actual operating point on this curve will depend on the system resistance. In this figure, the fan's operating point at "A" is flow (Q1) against pressure (P1).
- Two methods can be used to reduce air flow from Q1 to Q2:
 - **(Click once)** The first method is to restrict the air flow by partially closing a damper in the system. This action causes a new system performance curve (SC2) where the required pressure is greater for any given air flow. The fan will now operate at "B" to provide the reduced air flow Q2 against higher pressure P2.
 - **(Click once)** The second method to reduce air flow is by reducing the speed from N1 to N2, keeping the damper fully open. The fan would operate at "C" to provide the same Q2 air flow, but at a lower pressure P3. Thus, reducing the fan speed is a much more efficient method to decrease airflow since less power is required.

Fan Laws



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(BEE India, 2004)

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- The relationship between the fan speed on one hand, and the air flow, pressure and power requirements is described by fan laws.
- (Click once)** The speed is directly correlated with the air flow. For example, if the speed is increased by 10%, then the air flow also increases by 10%
- (Click once)** When the speed is increased then the pressure SP increases much more, as shown in the graph and formula. For example, a 10% speed increase results in a 21% pressure increase
- (Click once)** When the speed is increased then the power kW increases most, as shown in the graph and formula. For example, if the speed is increased by 10%, then the power requirement increases by 33%

 UNEP	Training Agenda: Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Introduction Types of fans and blowers Assessment of fans and blowers Energy efficiency opportunities 10 © UNEP 2006

- This section briefly describes about different types of fans and blowers.

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Types of fans • Centrifugal • Axial Types of blowers • Centrifugal • Positive displacement 11 © UNEP 2006

- There exist two main fan types. Centrifugal fans used a rotating impeller to move the air stream. Axial fans move the air stream along the axis of the fan.
- The centrifugal blower and the positive displacement blower are two main types of blowers.

These are described next.

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Centrifugal Fans • Rotating impeller increases air velocity • Air speed is converted to pressure • High pressures for harsh conditions • High temperatures • Moist/dirty air streams • Material handling • Categorized by blade shapes • Radial • Forward curved • Backward inclined 12 © UNEP 2006

- Centrifugal fans increase the speed of an air stream with a rotating impeller.
- The speed increases as the reaches the ends of the blades and is then converted to pressure.
- These fans are able to produce high pressures, which makes them suitable for harsh operating conditions, such as systems with high temperatures, moist or dirty air streams, and material handling.
- Centrifugal fans are categorized by their blade shapes.

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers 	Centrifugal Fans – Radial fans • Advantages • High pressure and temp • Simple design • High durability • Efficiency up to 75% • Large running clearances • Disadvantages • Suited for low/medium airflow rates only  (Canadian Blower) 13 © UNEP 2006

Radial fans, with flat blades

Advantages:

- Suitable for high static pressures (up to 1400 mmWC) and high temperatures
- Simple design allows custom build units for special applications
- Can operate at low air flows without vibration problems
- High durability
- Efficiencies up to 75%
- Have large running clearances, which is useful for airborne-solids (dust, wood chips and metal scraps) handling services

Disadvantages:

- Only suitable for low-medium airflow rates

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Centrifugal Fans – Forward curved • Advantages • Large air volumes against low pressure • Relative small size • Low noise level • Disadvantages • Not high pressure / harsh service • Difficult to adjust fan output • Careful driver selection • Low energy efficiency 55-65%  (Canadian Blower) 14 © UNEP 2006

Forward curved fans, with forward curved blades

Advantages:

- Can move large air volumes against relatively low pressure
- Relative small size
- Low noise level (due to low speed) and well suited for residential heating, ventilation, and air conditioning (HVAC) applications

Disadvantages:

- Only suitable for clean service applications but not for high pressure and harsh services
- Fan output is difficult to adjust accurately
- Driver must be selected carefully to avoid motor overload because power curve increases steadily with airflow
- Relatively low energy efficiency (55-65%)

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Centrifugal Fans - Backward-inclined • Advantages • Operates with changing static pressure • Suited for high flow and forced draft services • Efficiency >85% • Disadvantages • Not suited for dirty airstreams • Instability and erosion risk  (Canadian Blower) 15 © UNEP 2006

Backward inclined fan, with blades that tilt away from the direction of rotation: flat, curved, and airfoil

Advantages:

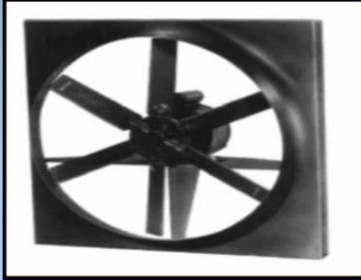
- Can operate with changing static pressure (as this does not overload the motor)
- Suitable when system behavior at high air flow is uncertain
- Suitable for forced-draft services
- Flat bladed fans are more robust
- Curved blades fans are more efficient (exceeding 85%)
- Thin air-foil blades fans are most efficient

Disadvantages:

- Not suitable for dirty air streams (as fan shape promotes accumulation of dust)
- Airfoil blades fans are less stable because of stall as they rely on the lift created by each blade
- Thin airfoil blades fans subject to erosion

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Axial Fans • Work like airplane propeller: • Blades create aerodynamic lift • Air is pressurized • Air moves along fan axis • Popular with industry: compact, low cost and light weight • Applications • Ventilation (requires reverse airflow) • Exhausts (dust, smoke, steam) 16 © UNEP 2006

- Axial fans move an air stream along the axis of the fan.
- The way these fans work can be compared to a propeller on an airplane: the fan blades generate an aerodynamic lift that pressurizes the air.
- They are popular with industry because they are inexpensive, compact and light.
- Although the fans are typically designed to generate flow in one direction, they can operate in the reverse direction too. This characteristic is useful when a space may require contaminated air to be exhausted or fresh air to be supplied. Axial fans are frequently used in exhaust applications where airborne particulate size is small, such as dust streams, smoke, and steam. Axial fans are also useful in ventilation applications that require the ability to generate reverse airflow.

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers 	Axial Fans – Propeller fans • Advantages • High airflow at low pressure • Little ductwork • Inexpensive • Suited for rooftop ventilation • Reverse flow • Disadvantages • Low energy efficiency • Noisy  (Fan air Company) 17 © UNEP 2006

Propeller fan (Figure 11)

Advantages:

- Generate high airflow rates at low pressures
- Not combined with extensive ductwork (because they generate little pressure)
- Inexpensive because of their simple construction
- Achieve maximum efficiency, near-free delivery, and are often used in rooftop ventilation applications
- Can generate flow in reverse direction, which is helpful in ventilation applications

Disadvantages:

- Relative low energy efficiency
- Comparatively noisy

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers 	Axial Fans – Tube axial fans • Advantages • High pressures to overcome duct losses • Suited for medium-pressure, high airflow rates • Quick acceleration • Space efficient • Disadvantages • Expensive • Moderate noise • Low energy efficiency 65%  (Canadian Blower) 18 © UNEP 2006

Tube-axial fan, essentially a propeller fan placed inside a cylinder (Figure 12)

Advantages:

- Higher pressures and better operating efficiencies than propeller fans
- Suited for medium-pressure, high airflow rate applications, e.g. ducted HVAC installations
- Can quickly accelerate to rated speed (because of their low rotating mass) and generate flow in reverse direction, which is useful in many ventilation applications
- Create sufficient pressure to overcome duct losses and are relatively space efficient, which is useful for exhaust applications

Disadvantages:

- Relatively expensive
- Moderate airflow noise
- Relatively low energy efficiency (65%)

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers 	Axial Fans – Vane axial fans • Advantages • Suited for medium/high pressures • Quick acceleration • Suited for direct motor shaft connection • Most energy efficient 85% • Disadvantages • Expensive  (Canadian Blower) 19 © UNEP 2006

Vane-axial fan (Figure 13)

Advantages:

- Suited for medium- to high-pressure applications (up to 500 mmWC), such as induced draft service for a boiler exhaust
- Can quickly accelerate to rated speed (because of their low rotating mass) and generate flow in reverse directions, which is useful in many ventilation applications
- Suited for direct connection to motor shafts
- Most energy efficient (up to 85% if equipped with airfoil fans and small clearances)

Disadvantages:

- Relatively expensive compared to propeller fans

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Blowers • Difference with fans • Much higher pressures <1.20 kg/cm² • Used to produce negative pressures for industrial vacuum systems • Types • Centrifugal blower • Positive displacement 20 © UNEP 2006

- Blowers can achieve much higher pressures than fans, as high as 1.20 kg/cm². They are also used to produce negative pressures for industrial vacuum systems. \
- The centrifugal blower and the positive displacement blower are two main types of blowers, which are described next

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers 	Centrifugal Blowers • Gear-driven impeller that accelerates air • Single and multi-stage blowers • Operate at 0.35-0.70 kg/cm² pressure • Airflow drops if system pressure rises  (Fan air Company) 21 © UNEP 2006

- Centrifugal blowers look more like centrifugal pumps than fans.
- The impeller is typically gear-driven and rotates as fast as 15,000 rpm. In multi-stage blowers, air is accelerated as it passes through each impeller. In single-stage blower, air does not take many turns, and hence it is more efficient.
- Centrifugal blowers typically operate against pressures of 0.35 to 0.70 kg/cm², but can achieve higher pressures.
- One characteristic is that airflow tends to drop drastically as system pressure increases, which can be a disadvantage in material conveying systems that depend on a steady air volume. Because of this, they are most often used in applications that are not prone to clogging.

 UNEP	Types of Fans & Blowers
Electrical Equipment Fans & Blowers Gerlap	Positive Displacement Blowers • Rotors trap air and push it through housing • Constant air volume regardless of system pressure • Suited for applications prone to clogging • Turn slower than centrifugal blowers • Belt-driven for speed changes 22 © UNEP 2006

Positive Displacement

- Positive displacement blowers have rotors, which "trap" air and push it through housing.
- These blowers provide a constant volume of air even if the system pressure varies. They are especially suitable for applications prone to clogging, since they can produce enough pressure (typically up to 1.25 kg/cm²) to blow clogged materials free.
- They turn much slower than centrifugal blowers (e.g. 3,600 rpm) and are often belt driven to facilitate speed changes.



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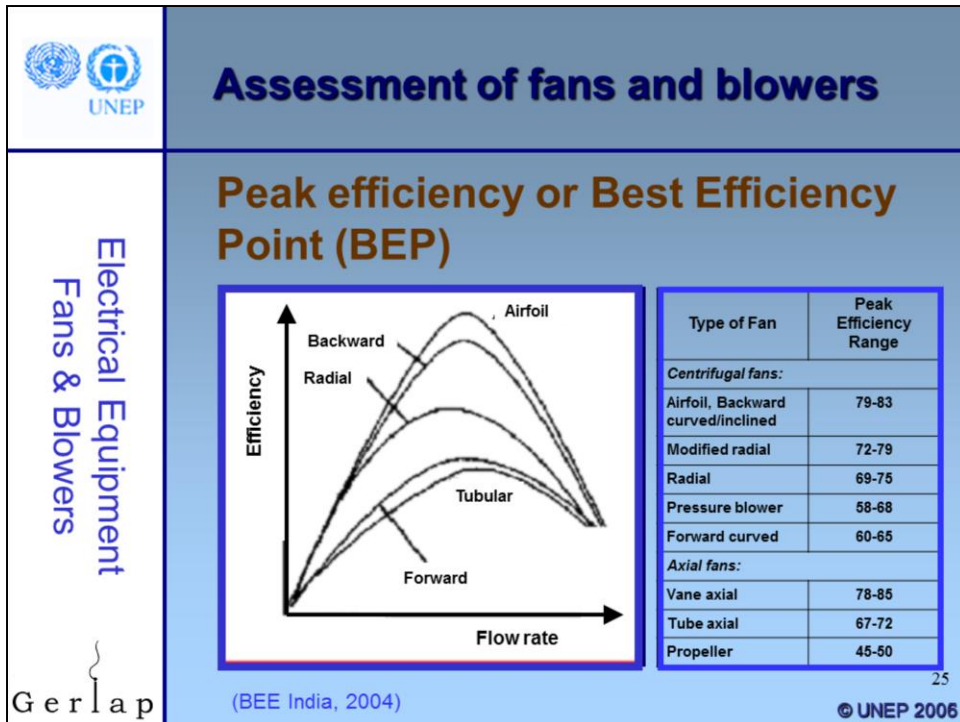
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 UNEP	Assessment of fans and blowers
Electrical Equipment Fans & Blowers Gerlap	Fan Efficiency and Performance • Fan efficiency: • Ratio of the power conveyed to air stream and power delivered by the motor to the fan • Depends on type of fan and impeller • Fan performance curve • Graph of different pressures and corresponding required power • Supplier by manufacturers 24 © UNEP 2006

• Fan efficiency is the ratio between the power transferred to the air stream and the power delivered by the motor to the fan. The power of the airflow is the product of the pressure and the flow, corrected for unit consistency. The fan efficiency depends on the type of fan and impeller.

• **(Click once)** We already discussed the fan performance curve earlier: a graph that shows the different pressures developed by the fan and the corresponding required power. The manufacturers normally provide these fan performance curves. Understanding this relationship is essential to designing, sourcing, and operating a fan system and is the key to optimum fan selection.



- As the flow rate increases, the efficiency increases to certain height (“peak efficiency”) and then decreases with further increasing flow rate. **(Point at the peak of one of the curves)**
- The peak efficiency is also called the Best Efficiency Point (BEP)
- The peak efficiency ranges for different types of centrifugal and axial fans are given in the Table.

 UNEP	Assessment of fans and blowers
Electrical Equipment Fans & Blowers Gerlap	Methodology – fan efficiency Before calculating fan efficiency • Measure operating parameters • Air velocity, pressure head, air stream temp, electrical motor input • Ensure that • Fan is operating at rated speed • Operations are at stable condition 26 © UNEP 2006

Before the fan efficiency can be calculated, a number of operating parameters must be measured, including air velocity, pressure head, temperature of air stream on the fan side and electrical motor kW input. In order to obtain correct operating figures it should be ensured that:

- Fan and its associated components are operating properly at its rated speed
- Operations are at stable condition i.e. steady temperature, densities, system resistance etc.

 UNEP	Assessment of fans and blowers
Electrical Equipment Fans & Blowers Gerlap	Methodology – fan efficiency Step 1: Calculate air/gas density $\text{Gas Density } (\gamma) = \frac{273 \times 1.293}{273 + t^{\circ}C}$ Step 2: Measure air velocity and calculate average $\text{Velocity } v, \text{ m/s} = \frac{C_p \times \sqrt{2 \times 9.81 \times \Delta p \times \gamma}}{\gamma}$ Step 3: Calculate the volumetric flow in the duct $\text{Volumetric flow}(Q), \text{ m}^3 / \text{sec} = \text{Velocity}, V (\text{m/sec}) \times \text{Area} (\text{m}^2)$ t = Temperature of air/gas at site condition C_p = Pitot tube constant, 0.85 (or) as given by the manufacturer Δp = Average differential pressure γ = Density of air or gas at test condition 27 © UNEP 2006

- The calculation of fan efficiency is explained in 5 steps.

(Click once) Step 1. The first step is to calculate the air or gas density using the given equation

(Click once) Step 2. The air velocity can be measured with a pitot tube and a manometer, or a flow sensor (differential pressure instrument), or an accurate anemometer. Calculate the average air velocity by taking number of velocity pressure readings across the cross-section of the duct using the given equation, where C_p is the pitot tube constant of 0.85 or as given by the manufacturer, and Δp is the average differential pressure.

(Click once) Step 3. Take the duct diameter (or the circumference from which the diameter can be estimated). Next, calculate the volume of air/gas in the duct by using the given formula

 UNEP	Assessment of fans and blowers
Electrical Equipment Fans & Blowers Gerlap	Methodology – fan efficiency Step 4: Measure the power drive of the motor Step 5: Calculate fan efficiency Fan mechanical efficiency $\text{Fan Mechanical Efficiency } (\eta_{\text{mechanical}}), \% = \frac{\text{Volume in m}^3 / \text{sec} * \Delta p (\text{total pressure}) \text{ in mmWC}}{102 * \text{power input to fan shaft in kW}} \times 100$ Fan static efficiency $\text{Fan Static Efficiency } (\eta_{\text{static}}), \% = \frac{\text{Volume in m}^3 / \text{sec} * \Delta p (\text{static pressure}) \text{ in mmWC}}{102 * \text{power input to fan shaft in kW}} \times 100$

• Step 4. The power of the drive motor (kW) can be measured by a load analyzer. This kW multiplied by motor efficiency gives the shaft power to the fan.

• **(Click once)** Step 5. Now the fan's mechanical and static efficiencies can be calculated using these equations

 UNEP	Assessment of fans and blowers
Electrical Equipment Fans & Blowers Gerlap	Difficulties in Performance Assessment • Non-availability of fan specification data • Difficulty in velocity measurement • Improper calibration of instruments • Variation of process parameters during tests 29 © UNEP 2006

In practice certain difficulties have to be faced when assessing the fan and blower performance, some of which are explained below:

•**Non-availability of fan specification data:** Fan specification data (see Worksheet 1) are essential to assess the fan performance. Most of the industries do not keep these data systematically or have none of these data available at all. In these cases, the percentage of fan loading with respect to flow or pressure can not be estimated satisfactorily. Fan specification data should be collected from the original equipment manufacturer (OEM) and kept on record.

•**Difficulty in velocity measurement:** Actual velocity measurement becomes a difficult task in fan performance assessment. In most cases the location of duct makes it difficult to take measurements and in other cases it becomes impossible to traverse the duct in both directions. If this is the case, then the velocity pressure can be measured in the center of the duct and corrected by multiplying it with a factor 0.9.

•**Improper calibration of the pitot tube, manometer, anemometer & measuring instruments:** All instruments and other power measuring instruments should be calibrated correctly to avoid an incorrect assessment of fans and blowers. Assessment should not be carried out by applying correction factors to compensate for this.

•***Variation of process parameters during tests:*** If there is a large variation of process parameters measured during test periods, then the performance assessment becomes unreliable.



Training Agenda: Fans & Blowers

Electrical Equipment
Fans & Blowers

Gerlap

Introduction

Types of fans and blowers

Assessment of fans and blowers

Energy efficiency opportunities

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 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers 	1. Choose the right fan 2. Reduce the system resistance 3. Operate close to BEP 4. Maintain fans regularly 5. Control the fan air flow 31 © UNEP 2006

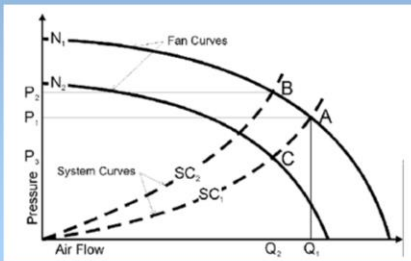
There are five main areas for energy conservation for fans which we will discuss on the next slides.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers 	1. Choose the Right Fan • Considerations for fan selection • Noise • Rotational speed • Air stream characteristics • Temperature range • Variations in operating conditions • Space constraints and system layout • Purchase/operating costs and operating life • “Systems approach” most important! 32 © UNEP 2006

- Important considerations when selecting a fan are:
 - Noise
 - Rotational speed
 - Air stream characteristics
 - Temperature range
 - Variations in operating conditions
 - Space constraints and system layout
 - Purchase costs, operating costs (determined by efficiency and maintenance), and operating life
- But as a general rule it is important to know that to effectively improve the performance of fan systems, designers and operators must understand how other system components function as well. The “systems approach” requires knowing the interaction between fans, the equipment that supports fan operation, and the components that are served by fans. The use of a “systems approach” in the fan selection process will result in a quieter, more efficient, and more reliable system.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers 	1. Choose the Right Fan • Avoid buying oversized fans • Do not operate at Best Efficiency Point • Risk of unstable operation • Excess flow energy • High airflow noise • Stress on fan and system 33 © UNEP 2006

• A common problem is that companies purchase oversized fans for their service requirements. They will not operate at their best efficiency point (BEP) and in extreme cases these fans may operate in an unstable manner because of the point of operation on the fan airflow-pressure curve. Oversized fans generate excess flow energy, resulting in high airflow noise and increased stress on the fan and the system. Consequently, oversized fans not only cost more to purchase and to operate, they create avoidable system performance problems. Possible solutions include, amongst other replacing the fan, replacing the motor, or introducing a variable speed drive motor.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers 	2. Reduce the System Resistance • Increased system resistance reduces fan efficiency • Check periodically • Check after system modifications • Reduce where possible  (BEE India, 2004) 34 © UNEP 2006

- The system resistance curve and the fan curve were explained earlier. The fan operates at a point where the system resistance curve and the fan curve intersects.
- The system resistance has a major role in determining the performance and efficiency of a fan. In the figure, if the system resistance is increased then the operating point moves from A to B. The result is that the air flow of the fan reduces, and thus the fan efficiency.
- The system resistance changes
 - Marginally by the formation of the coatings / erosion of the lining in the ducts
 - Drastically, in some cases, due to the change of equipment, duct modifications
- Hence, the system resistance has to be periodically checked, more so when modifications are introduced and action taken accordingly to reduce the system resistance, for efficient operation of the fan.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers 	3. Operate Close to BEP • Best Efficiency Point = maximum efficiency • Normally close to rated fan capacity • Deviation from BEP results in inefficiency and energy loss 35 © UNEP 2006

- It is earlier described that the fan efficiency increases as the flow increases to certain point and thereafter it decreases. The point at which maximum efficiency is obtained is called the peak efficiency or “Best Efficiency Point” (BEP).
- Normally it is closer to the rated capacity of the fan at a particular designed speed and system resistance.
- Deviation from the BEP will result in increased loss and inefficiency.

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	4. Maintain Fans Regularly • Periodic inspection of all system components • Bearing lubrication and replacement • Belt tightening and replacement • Motor repair or replacement • Fan cleaning 36 © UNEP 2006

Regular maintenance of fans is important to maintain their performance levels. Maintenance activities include:

- Periodic inspection of all system components
- Bearing lubrication and replacement
- Belt tightening and replacement
- Motor repair or replacement
- Fan cleaning

5. Control the Fan Air flow

- a) Pulley change
- b) Dampers
- c) Inlet guide vanes
- d) Variable pitch fans
- e) Variable speed drives (VSD)
- f) Multiple speed drive
- g) Disc throttle
- h) Operating fans in parallel
- i) Operating fans in series



UNEP

Energy Efficiency Opportunities

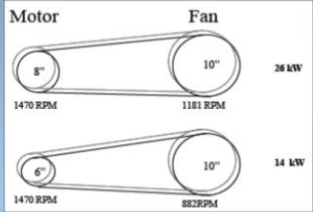
Electrical Equipment
Fans & Blowers

Gerlap

5. Control the Fan Air flow

a) Pulley change: reduce motor/drive pulley size

- **Advantages**
 - Permanent speed decrease
 - Real energy reduction
- **Disadvantages**
 - Fan must handle capacity change
 - Only applicable if V-belt system or motor



(BEE India, 2004)

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Pulley change: reduces the motor / drive pulley size

(Click once) Advantages:

- Permanent speed decrease
- Real energy reduction (**Explain the figure:** a 2 inch reduction in pulley, from 8 inch to 6 inch, results in 12 kW savings)

Disadvantages:

- Fan must be able to handle capacity change
- Fan must be driven by V-belt system or motor

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow b) Dampers: reduce flow and increase upstream pressure • Advantages • Inexpensive • Easy to install • Disadvantages • Limited adjustment • Reduce flow but not energy consumption • Higher operating and maintenance costs 39 © UNEP 2006

Dampers: reduce the amount of flow and increases the upstream pressure, which reduces fan output

(Click once)

Advantages:

- Inexpensive
- Easy to install

Disadvantages:

- Provide a limited amount of adjustment
- Reduce the flow but not the energy consumption
- Higher operating and maintenance costs

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow c) Inlet guide vanes • Create swirls in fan direction • Reduce angle air and fan blades • Lowering fan load, pressure, air flow • Advantages • Improve efficiency: reduced load and airflow • Cost effective at 80-100% of full air flow • Disadvantage • Less efficient at <80% of full air flow 40 © UNEP 2006

Inlet Guide vanes:

• Inlet guide vanes: create swirls in the fan direction thereby lessening the angle between incoming air and fan blades, and thus lowering fan load, pressure and airflow

(Click once)

Advantages:

- Improve fan efficiency because both fan load and delivered airflow are reduced
- Cost effective at airflows between 80-100% of full flow

Disadvantages:

- Less efficient at airflows lower than 80% of full flow

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow d) Variable pitch fans: changes angle incoming airflow and blades • Advantages • High efficiency at range of operating conditions • No resonance problems • No stall problems at different flows • Disadvantages • Applicable to axial fans only • Risk of fouling problems • Reduced efficiency at low loads 41 © UNEP 2006

Variable-pitch fans

Change the angle between incoming airflow and the blade by tilting the fan blades, thereby reducing both the motor load and airflow

(Click once)

Advantages:

- Can keep fan efficiency high over a range of operating conditions.
- Avoid resonance problems as normal operating speed is maintained
- Can operate from a no-flow to a full-flow condition without stall problems

Disadvantages:

- Applicable to some axial fan types only
- Fouling problems if contaminants accumulate in the mechanical actuator that controls the blades
- Operating at low loads for long periods reduces the power factor and motor efficiency, thus losing efficiency advantages and risking low power factor charge from the utility

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow e) Variable speed drives (VSDs): reduce fan speed and air flow • Two types • Mechanical VSDs • Electrical VSDs (including VFDs) • Advantages • Most improved and efficient speed control • Speed adjustments over continuous range • Disadvantage: high costs 42 © UNEP 2006

Variable Speed Drive (VSD): reducing the speed of the fan to meet reduced flow requirements

- Mechanical VSDs: hydraulic clutches, fluid couplings, and adjustable belts and pulleys
- Electrical VSDs: eddy current clutches, wound-rotor motor controllers, and variable frequency drives (VFDs: change motor's rotational speed by adjusting electrical frequency of power supplied)

(Click once)

Advantages:

- Most improved and efficient flow control
- Allow fan speed adjustments over a continuous range

Disadvantages:

- Mechanical VSDs have fouling problems
- Investment costs can be a barrier

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow e) Variable frequency drives • Change motor's rotational speed by adjusting electrical frequency of power • Advantages • Effective and easy flow control • Improved efficiency over wide operating range • Can be retrofitted to existing motors • Compactness • No fouling problems • Reduced energy losses and costs 43 © UNEP 2006

Variable frequency drives (VFDs)

- VFDs are the most commonly used type of electrical VSD used
- Change motor's rotational speed by adjusting electrical frequency of power supplied

(Click once)

Advantages for VFDs specifically:

- Effective and easy flow control
- Improve fan operating efficiency over a wide range of operating conditions
- Can be retrofitted to existing motors
- Compactness
- No fouling problems
- Reduce energy losses and costs by lowering overall system flow

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow f) Multiple speed drive • Changes fan speed from one speed to other speed • Advantages • Efficient control of flow • Suitable if only 2 speeds required • Disadvantages • Need to jump from speed to speed • High investment costs 44 © UNEP 2006

Multiple speed drive

(Click once)

Advantages

- Efficient control of flow
- Suitable if only two fixed speeds are required

Disadvantages

- Need to jump from speed to speed
- Investment costs can be a barrier

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow g) Disc throttle: Sliding throttle that changes width of impeller exposed to air stream • Advantages • Simple design • Disadvantages • Feasible in some applications only 45 © UNEP 2006

Disc throttle: a sliding throttle that changes the width of the impeller that is exposed to the air stream

(Click once)

Advantages:

- Simple design

Disadvantages:

- Feasible in some applications only

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers Gerlap	5. Control the Fan Air flow h) Operate more fans in parallel (instead of one large fan) • Advantages • High efficiencies at varying demand • Risk of downtime avoided • Less expensive and better performance than one large fan • Can be equipped with other flow controls • Disadvantages • Only suited for low resistance system 46 © UNEP 2006

Operate fans in parallel:

- two or more fans in parallel instead of one large one

(Click once)

Advantages:

- High efficiencies across wide variations in system demand
- Redundancy to mitigate the risk of downtime because of failure or unexpected maintenance
- Two smaller fans are less expensive and offer better performance than one relatively large one
- Can be equipped with other flow controls to increase flexibility and reliability

Disadvantages:

- Should only be used when the fans can operate in a low resistance almost in a free delivery condition

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers 	5. Control the Fan Air flow i) Operate fans in series • Advantages • Lower average duct pressure • Less noise • Lower structural / electrical support required • Disadvantages • Not suited for low resistance systems 47 © UNEP 2006

Operate fans in series:

- using multiple fans in a push-pull arrangement

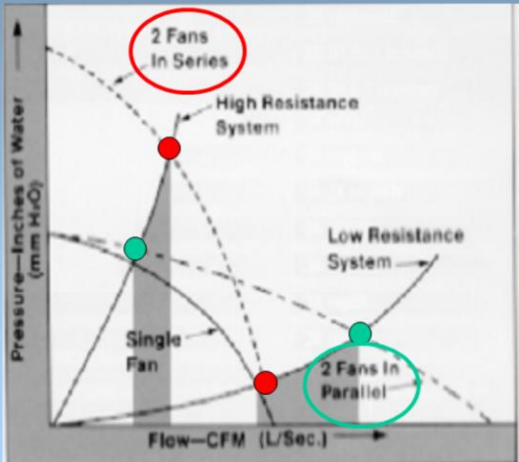
(Click once)

Advantages:

- Lower average duct pressure
- Lower noise generation
- Lower structural and electrical support requirements
- Suited for systems with long ducts, large pressure drops across system components, or high resistances

Disadvantages:

- Not suited for low resistance systems

 UNEP	Energy Efficiency Opportunities
Electrical Equipment Fans & Blowers 	5. Controlling the Fan Air Flow Comparing Fans in Parallel and Series  48 © UNEP 2006

This figure compares the fan curves of fans in series in fans in parallel

(Click once) The fan curve of two fans in series starts high but drops rapidly. At the point where the fan curve meets the High Resistance System curve, i.e. the operating point, the flow is much higher than that of a single fan. In other words, the efficiency gain is significant. But at the operating point in a Low Resistance System, the flow is low and practically the same as that of a single fan. This is why fans in series are not suited for low resistance systems

(Click once) For fans in parallel we see exactly the opposite. The fan curve of two fans in parallel does not start high but drops much slower. In a High Resistance System the operating point is only slightly higher than that of a single fan. In other words, the efficiency gain in a High Resistance System is minimal. But in a Low Resistance System the operating point of two fans in parallel is relatively high. This is why fans in parallel are suited for low resistance systems.



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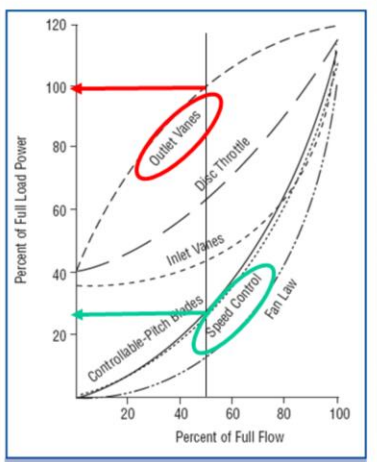
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5. Controlling the Fan Air Flow

Comparing the impact of different types of flow control on power use



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This figure shows how some types of flow control are more effective in reducing power consumption than others.

For example, let's compare the use of outlet vanes and speed control

- (Click once)** Outlet vanes reduce the flow of air but not the fan speed, and therefore not the power consumption. At a 50% of full air flow, the power consumption is still close to 100% of full load power.
- (Click once)** Speed control with VSDs reduces the flow by reducing the speed and therefore the power consumption. At a 50% of full air flow, the power consumption has reduced to about 25% of full load power.



Training Session on Energy Equipment

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**THANK YOU
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